

BALRAJ PARSI

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PROFESSIONAL SUMMARY

Backend Software Engineer with 6+ years specialized in high-throughput Kafka pipelines, cloud-native data systems, and production microservice ownership across Fortune 500 companies including Capital One, Microsoft, and Amazon. Delivered event-driven pipelines processing 80K-100K daily record changes in a regulated banking environment, supported a deployment platform operating across 6M+ EC2 hosts, and maintained backend services across 5+ global regions. Known for fast incident resolution, end-to-end delivery without dedicated QA, and taking systems from design to production ownership.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, C#, KQL

Frameworks & Libraries: Spring Boot, Node.js, JUnit, Mockito, Selenium

Databases: DynamoDB, Aurora DB (PostgreSQL), MySQL, Azure Cosmos DB, Amazon RDS

Messaging & Streaming: Apache Kafka, Kafka Streams, AWS SQS, AWS SNS, ActiveMQ

AWS Services: EC2, S3, Lambda, CloudFormation, Route 53, Secrets Manager, CloudWatch, CodeDeploy, DynamoDB Streams, ECR

Azure: Azure DevOps, ARM Templates, AKS, EV2, Azure Monitor, KQL / Kusto

CI/CD & DevOps: Jenkins, JFrog Artifactory, SonarQube, Mend, Docker, GitHub Actions

Practices: Microservices, RESTful APIs, OAuth2, Event-Driven Architecture, Agile / Scrum, On-Call Production Ownership

AI & ML: Generative AI, Large Language Models (LLMs), LLM API Integration, Prompt Engineering (COSTAR, ACR), Machine Learning Foundations

AI-Assisted Engineering: GitHub Copilot, Claude Code, Agentic Development Workflows, AI-Assisted Prototyping

EXPERIENCE

Backend Software Engineer | Capital One

Jun 2025 - Present · Richmond, VA

Bank Tech - Small Business Bank (SBB) Modernization | Java, Spring Boot, Kafka Streams, DynamoDB, Aurora DB, AWS, Jenkins

- Eliminate enterprise-level PII data duplication across legacy Systematics and modern RA systems by operating Kafka Streams pipelines that process 80K-100K customer record changes nightly in a regulated banking environment.
- Design and maintain a bidirectional Business Core to Systematics sync pipeline using event-driven Lambda triggered by MQ messages, enabling real-time data consistency across 2 core banking systems.
- Migrated Aurora DB to DynamoDB by redesigning data models for key-value access patterns, enabling single-digit millisecond reads and eliminating relational query overhead for primary SBB lookup flows.
- Eliminated manual 90-day certificate rotation risk by migrating from .jks-based auth to Turing OAuth2 across the SBB microservice fleet, improving security posture and reducing operational overhead.
- Diagnosed and resolved a VP-escalated production incident in under 40 minutes, containing impact to 9 of 100K+ daily transactions by identifying a null-overwrite edge case in the customer address update flow.
- Implemented DynamoDB Streams and Lambda-based audit logging for customer PII changes, meeting security compliance requirements and providing full traceability across the platform.
- Own 4 Java microservices across 15 repositories, shipping 2-3 production deployments weekly on a 3-person team with no dedicated QA, using GitHub Copilot and Claude Code agents in IntelliJ to accelerate development and code review.

Software Engineer | Microsoft

Jan 2024 - May 2025 · Redmond, WA

Azure Resource Graph (ARG) Backend Services | Java, C#, Azure DevOps, KQL, ARM Templates, AKS, Azure Monitor

- Owned backend Java services for Azure Resource Graph, maintaining availability across 5+ global regions and supporting query workloads for millions of Azure resource records.
- Executed phased GTM traffic rebalancing across regional endpoints, reducing failed deployments during rollouts and enabling zero-downtime releases for ARG backend services.
- Migrated Azure DevOps CI/CD pipelines from Classic to YAML for the ARG backend, enabling code-reviewed pipeline changes via pull requests and standardizing delivery across the team.
- Built PowerShell and C# validation scripts, accelerated with GitHub Copilot, that caught configuration drift and data inconsistencies pre-promotion, reducing deployment failures and shortening the debugging cycle.

- Built KQL-based monitoring dashboards and alerting in Azure Monitor for backend service health, reducing mean time to detect production issues.

Software Development Engineer | Amazon Web Services

Jun 2022 - Dec 2023 · Seattle, WA

Apollo Deployment Platform | Node.js, Java, Spring Boot, AWS (EC2, S3, DynamoDB), REST APIs, Docker

- Supported a deployment platform operating across 6M+ EC2 hosts at Amazon, contributing Node.js microservices that handled software delivery to S3, DynamoDB, and core Amazon Retail systems.
- Designed RESTful APIs for the Apollo Import transport pipeline and authored unit tests for the Apollo Client, reducing integration failures in the deployment transport layer.
- Built a Node.js load-balancing service that distributed deployment requests across Apollo agents, improving throughput and reducing queue latency for high-volume deployment events.
- Automated container image cleanup using AWS Lambda and CloudWatch Events, reducing stale image accumulation and keeping ECR repositories within retention policy.
- Contributed to migrating on-premises workloads onto Apollo's AWS infrastructure, redesigning service boundaries and data flows to support cloud-native deployment patterns.

Software Engineer | HCL Technologies

Jul 2019 - Aug 2021 · Chennai, India

Cisco CA-BOLT Project | Python, REST APIs, Cisco NXOS / iOS / ACI / VPN / AnyConnect, Selenium, Paramiko

- Developed 350+ Python network modules for Cisco CA-BOLT using REST APIs and SSH-based device communication via Paramiko, automating configuration and management across NXOS, iOS, ACI, VPN, and AnyConnect product lines.
- Built Selenium-based automated regression suites covering 200+ test cases across Cisco device categories, reducing manual testing effort by 40% and cutting cycle time for network software updates.
- Wrote Python scripts using Paramiko and subprocess to automate log collection, SSH health checks, and alert generation across the Cisco device fleet, eliminating recurring manual toil for the operations team.

PROJECTS

- **Resume Chatbot:** Built a conversational interface over my resume using an LLM API and serverless functions (vanilla JS frontend), deployed on Vercel as an interactive alternative to a static PDF. [Live demo](#)
- **Portfolio Site (balrajparasi.github.io):** Designed and deployed a personal portfolio on GitHub Pages presenting projects, experience, and technical writing.

EDUCATION

Executive MS, Artificial Intelligence

Oct 2025 - Present

University of the Cumberland

MS, Computer Science (Data Science)

GPA: 3.83 · Aug 2021 - Dec 2022

University of Cincinnati

B.Tech, Electronics and Communication Engineering

Aug 2015 – Apr 2019

Anurag Group of Institutions, JNTUH, India